

1. Analytical Objective & Problem Framing

The objective of challenge E3 is to quantify and benchmark creator audience engagement across all recognized publishing channels in the posts table. The analysis requires computing the arithmetic mean of individual engagement vectors (likes, shares, comments) and determining which platform drives the highest combined average interaction volume.

2. Target Relational Schema

All data resides within the **posts** table (12,000 total records). The schema contains four interaction attributes: platform (TEXT), likes (INTEGER, 1,814 missing), shares (INTEGER, 0 missing), and comments (INTEGER, 0 missing). No inter-table joins are required, keeping computational complexity minimal.

3. Step-by-Step SQL Architecture

- Step 1 - Channel Filtering (WHERE platform IS NOT NULL):** The official question specifies: *"Calculate the average likes, shares, and comments for each platform."* Posts with missing platform strings cannot be attributed to any channel and are excluded, leaving exactly 10,216 valid posts.
- Step 2 - Channel Grouping (GROUP BY platform):** The 10,216 posts are partitioned into five discrete platform clusters: Facebook (2,074), YouTube (2,073), Twitter (2,049), Reddit (2,031), and Instagram (1,989).
- Step 3 - Individual Interaction Means:** Evaluated via ROUND(AVG(metric), 2). In ANSI SQL, AVG() ignores NULL values in denominator and numerator, yielding the authentic empirical mean of observed interactions.
- Step 4 - Total Engagement Formulation:** Total interactions per post represent likes + shares + comments. Using COALESCE(likes, 0) ensures that posts with unrecorded likes still contribute their observed shares and comments.
- Step 5 - Ranking:** ORDER BY avg_total_engagement DESC sorts platforms strictly from highest to lowest.

4. Methodological Verification of NULL Handling

To ensure complete transparency, we audited the three standard SQL approaches for computing total engagement across platforms:

Platform	Post Count	Method A: COALESCE(likes,0) (Official Query)	Method B: Complete Cases (likes + shares + comments)	Method C: Sum of Means (AVG(L)+AVG(S)+AVG(C))
Instagram	1,989	3,669.38 (#1)	4,040.02 (#1)	4,041.57 (#1)
Reddit	2,031	3,647.47 (#2)	4,003.94 (#3)	4,001.52 (#3)
YouTube	2,073	3,638.03 (#3)	4,031.84 (#2)	4,033.98 (#2)
Facebook	2,074	3,631.00 (#4)	4,015.99 (#4)	4,019.97 (#4)
Twitter	2,049	3,563.75 (#5)	3,951.91 (#5)	3,949.21 (#5)

Conclusion on Invariance: Regardless of whether missing likes are treated as 0 (Method A), filtered out via complete-case analysis (Method B), or aggregated as independent vectors (Method C), **Instagram is unequivocally the #1 engagement channel across all definitions.**